PhD Lecture in Behavioral Asset Pricing: Memory and Markets

A New Psychological Primitive for Beliefs and Prices

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Why Memory?

- **Beliefs are the central object of asset pricing.**
  - What investors expect about future returns drives prices, volume, and bubbles
  - Standard models assume investors *use* the full history of information efficiently
- **But human memory is not a faithful archive.**
  - Recall is *reconstructive* and *cue-dependent*: what comes to mind depends on the current context
  - Psychologists have known this for 50 years (Tulving & Thomson, 1973; Kahana, 2012)
- **A new literature asks: does this matter for markets?**
  - If beliefs are formed from *recalled* (not actual) experiences, mispricing follows
  - Memory is distinct from — and upstream of — sentiment, attention, and extrapolation

This lecture

Four recent papers make memory a first-class primitive in behavioral finance: from the lab, to individual investors, to the cross-section of stocks, to market prices.

Roadmap

1. **Psychology foundation** — associative memory and cue-dependent retrieval
2. **Theory** — from recalled experiences to biased beliefs
3. **Causal lab evidence** — Enke, Schwerter & Zimmermann (2024, *JFE*)
4. **Field evidence: individual investors** — Jiang, Liu, Peng & Yan (2025, *QJE*)
5. **Marketwide evidence** — Charles & Sui (2024, working paper)
6. **Memory moves prices** — Charles (2024, *RFS*)
7. **Synthesis, open questions, and a research agenda**

Guiding question

Does memory provide a *microfoundation* for the beliefs and prices we see — overreaction, extrapolation, comovement, crisis echoes?

Associative Memory: The Psychology

- **Memory is not a tape recorder.** Recall is *cue-dependent*: a present stimulus (the *cue*) retrieves past episodes that resemble it (Tulving & Thomson, 1973; Kahana, 2012)
- **Four robust laws of cued recall** (Kahana, 2012):
  - *Similarity* — episodes similar to the cue are retrieved more easily
  - *Interference* — a focal episode is harder to retrieve when many similar episodes compete
  - *Contiguity* — episodes experienced close in time tend to be co-retrieved
  - *Recency* — recent episodes are more available
- **Key implication for finance:**
  - Today's market return / news / context acts as a *cue*
  - It asymmetrically retrieves *congruent* past experiences
  - Beliefs are built from this *biased sample* of memory, not from the full history

From Memory to Beliefs: A Two-Step Mechanism

Step 1 — Cued recall.

Today's context κ_t retrieves past experiences $\{e\}$ with probability

$$r(e, \kappa_t) \;=\; \frac{S(e, \kappa_t)}{\sum_{e'} S(e', \kappa_t)}, \qquad S(e, \kappa_t) \in [0, 1]$$

where S is a similarity kernel (e.g. cosine).

Step 2 — Simulation.

The investor averages retrieved experiences to *simulate* a future return distribution, then forecasts from it.

Memory-based expectation

$$\mathbb{E}^{\text{mem}}[r_{t+1}] \;=\; \sum_{e} r(e, \kappa_t)\, r(e)$$

Beliefs are a *recall-weighted* average of past returns — not a time-weighted one.

Foundations: Bordalo, Burro, Coffman, Gennaioli & Shleifer (2025, *RES*); Wachter & Kahana (2024); Gennaioli & Shleifer (2018).

Punchline: *extrapolation* and *recency emerge* from this mechanism — they are not assumed as primitives.

How Memory Differs from Neighbouring Concepts

- **vs. Extrapolation** (Greenwood & Shleifer, 2014; Barberis et al., 2018): extrapolation is a *reduced-form* fact; memory is its *mechanism*
- **vs. Experience effects** (Malmendier & Nagel, 2011, 2016): experience weights past returns by *fixed* exponential decay; memory weights them by *context-dependent* similarity — so recall *spikes* during crises
- **vs. Sentiment** (Baker & Wurgler, 2006): sentiment is a market-wide mood; memory is a within-inferior cognitive process that *produces* the belief input
- **vs. Attention** (Barber & Odean, 2008; Hirshleifer, Lim & Teoh, 2009): attention selects *which external* stimuli to process; memory generates *internal* cues from the past

Position in the literature

Memory sits *upstream* — it shapes the beliefs that extrapolation, sentiment, and attention then act on.

Enke, Schwerter & Zimmermann (2024, JFE): Design

- **Question:** does associative recall *cause* overreaction in expectations and prices?
- **Setup:** three pre-registered lab experiments
  - Subjects observe signals about a binary state $\theta \in \{G, B\}$ over two periods
  - First period: $k \in \{0, \ldots, 4\}$ signals. Second period: one signal s_{k+1}
  - Each signal is embedded in a *context* (a background image / narrative)
- **Manipulation (within-subject):**
  - **Cue:** signals congruent in sign share the same context $\Rightarrow$ cue facilitates recall
  - **NoCue:** contexts never repeat $\Rightarrow$ no associative retrieval
- **Identification:** z = number of first-period signals *congruent* with s_{k+1} is random; the Bayesian benchmark is a coefficient of *exactly 1* on s_{k+1}
- **Three experiments:** Beliefs (incentivized), Recall (surprise elicitation), Betting Market (parimutuel, groups of 3)

Enke et al. (2024): Results

Beliefs overreact only when memory is cued.

- Coefficient on s_{k+1}: **1.20** in Cue vs. ≈**1.0** in NoCue
- Treatment × signal interaction: **0.61***** (s.e. 0.15) — >80% larger response
- Rises monotonically in z (# congruent signals)
- Null when $z = 0$ — rules out "better recall in general"

Mechanism confirmed.
Subjects retrieve *congruent* first-period signals in Cue, not incongruent ones.

Markets do not attenuate

Market-price overreaction (**0.65*****) ≈ individual-belief overreaction (**0.61*****). Self-selection fails to dampen prices: the lab market *aggregates* the bias rather than correcting it.

Takeaway: a clean causal link from associative recall → belief overreaction → price overreaction. Narratives move prices through *retrieval*, not through information.

Jiang, Liu, Peng & Yan (2025, QJE): Design

- **Question:** do investors form return expectations by selectively *recalling* past experiences — and do recalled (not actual) experiences drive beliefs and trading?
- **Data:**
  - Nationally representative survey of ∼**17,000 Chinese retail investors**, late 2021
  - Survey rolled out in three waves over six weeks ⇒ plausibly exogenous variation in daily market returns (the *cue*)
  - Merged subsample (∼**5,154**) with administrative brokerage records
- **Two recall elicitations:**
  - **FreeRecall:** "recall a market episode that first comes to mind" (Neutral / Happy / Painful)
  - **ProbedRecall:** recall own return over horizons from yesterday to 5 years
- **Identification:** survey-day market return as the cue; respondents instructed not to consult external sources
- **Validity:** recalled vs. actual market-return correlation = 0.57

Jiang et al. (2025): Results

- **Cued recall.** +1pp survey-day market return ⇒ +**2.5–4.1pp** recalled episode return; +**52bps** recalled own return (yesterday), +**113bps** (past month) — *after controlling for actual own return*
- **Recall → beliefs.** Moving recalled past-month own return from 25th to 75th percentile raises 1-month expectation by **0.7pp** and 1-year expectation by **1.8pp**
- **Recalled dominates actual.** In horse races, recalled experiences *out-explain* actual experiences; controlling for recalls *weakens* the actual-experience–expectations link
- **Horizon dependence.** 1-year forecasts weight recalled 1-year returns; 1-month forecasts weight recalled 1-month returns
- **Memory → trading.** +1pp memory-induced own-return expectation raises net buying by **71 RMB** (survey day) and **1,141 RMB** (next month) — far larger than raw expectations

Punchline

"A rising market prompts investors to recall their past experiences more positively, leading to more optimistic forecasts" — *extrapolation as a memory artifact*.

Charles & Sui (2024): Marketwide Memory

- **Question:** can the laws of associative memory, applied at scale to the cross-section of stocks over decades, explain investor beliefs and trading?
- **The measure:** for each stock-month, reconstruct a *recall probability* over the past 180 months using

$$r(e, \kappa_t) = \frac{S(e, \kappa_t)}{\sum_{e'} S(e', \kappa_t)}, \qquad S(e, \kappa_t) = \tfrac{1}{2}\left[\cos(e, \kappa_t) + 1\right]$$

  where e is a past month encoded by stock-level, firm-level, and macro features
- **Key features of the measure:**
  - *Similarity* and *interference* both operate — interference makes recall *context-dependent*
  - *Recency emerges endogenously* — no exponential decay assumed
  - *Crisis echoes*: COVID-2020 cues 2008; 2008 cues 2001; 1979 cues 1973 — patterns pure extrapolation *cannot* produce
- **Scope:** CRSP stocks listed $\geq$15 years, 1966–2021

Charles & Sui (2024): Validation and Findings

Validates against real recall.

- Correlation **0.88** (rank 0.79) with mentions in earnings-call transcripts (Refinitiv, 2001–2021)
- 1pp higher recall prob $\Rightarrow$ **67pp** higher mention probability; exponential-decay weight becomes insignificant

Explains beliefs.

- Corr **0.67** with Gallup bullish-minus-bearish; 1 SD $\Rightarrow$ **14.5pp** more bullish investors
- Survives controls for past-12m and 5-year exp-weighted return

Explains higher moments.

- Memory-based volatility tracks option-implied vol (1 SD $\Rightarrow$ 59% of SD); time-series version explains VIX

Explains trading.

- For the same realized gain, raising recall prob 0$\rightarrow$10% raises repurchase likelihood by **0.11pp** ($\sim$22% of unconditional rate)

Distinction

Recency is not assumed — it *emerges* from similarity and interference. Crisis-echo spikes are the signature of memory, not extrapolation.

Charles (2024, RFS): Memory Moves Markets

- **Question:** can *internally generated* attention — driven by associative memory links between firms — distort stock prices, separately from any external attention stimulus?
- **The association:** two firms that announced earnings on the *same day* last quarter become *associated in memory* (shared encoding context)
- **The cue:** this quarter, when only firm A announces, the event cues recall of firm B $\Rightarrow$ internally generated attention to B $\Rightarrow$ retail net buying (Barber & Odean, 2008) $\Rightarrow$ price pressure
- **Identification:** Noh, So & Verdi (2021) "pattern firms" — firms following rigid announcement-timing rules (e.g. first Thursday of month) for $\geq$3 years. Calendar rotations exogenously shift announcement dates, creating plausibly random overlaps
- **Sample:** 2005–2019, $\sim$16.5M firm-days; CRSP/Compustat/I/B/E/S; retail order imbalance from TAQ
- **Controls:** own-EA dummies, 8-K filings, RavenPack sentiment/volume, day FE, firm-year-quarter FE; cueing firms in *different* TNIC industry, not vertically related, no shared analysts

Charles (2024): Price Dynamics

Cue → price.

- Baseline: cue ⇒ **+4bps** $[t, t+1]$, **+8bps** $[t, t+10]$ (full controls, 1% sig.)
- **Builds over ∼2 weeks, then fully reverses over ∼3 weeks** — zero by ∼25 trading days
- Inconsistent with fundamental information; signature of attention-driven mispricing

Retail-driven.

- Retail order imbalance positive and significant over $[t, t+10]$
- Effect concentrated in **small cued firms**
- Stronger post-2015 (Robinhood era): trading strategy alpha **+69.3bps** over $[t, t+10]$, ∼16% annualized

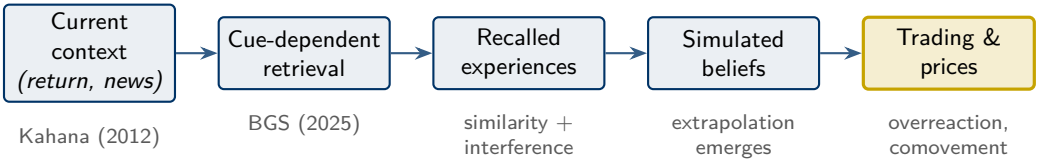
Internal vs external attention

Memory associations inside an investor's mind are an *internal* source of attention, distinct from all external sources previously studied (news, volume, extreme returns).

Mechanism checks: effect stronger when encoding-day volume high; *extremely negative* co-surprises yield *negative* response (similarity). Interference and contiguity directionally consistent.

Limits to arbitrage: small stocks, shorting costs — exactly where mispricing survives.

A Unified Framework: From Context to Prices



Four papers as building blocks:

- **ESZ (2024):** causal link, lab
- **JLPY (2025):** recalled > actual, field
- **Charles & Sui (2024):** scale to cross-section
- **Charles (2024):** memory moves prices

Memory is the *mechanism*; extrapolation, comovement, and crisis echoes are the *consequences*.

What Memory Explains — and What It Doesn't (Yet)

Explains

- Overreaction to cued signals (ESZ)
- Return extrapolation as a recall artifact (JLPY)
- Recency *and* crisis-echo spikes (Charles & Sui)
- Repurchase and disposition-style trading
- Internally-generated comovement & price pressure (Charles)
- Why lab designs (no memory) find *under*reaction while field finds *over*reaction

Doesn't (yet)

- Aggregation: how individual recall becomes market price
- Institutional / sophisticated investors
- Long-horizon dynamics and bubbles/crashes
- Trauma vs. contextual memory
- Interaction with sentiment and limits to arbitrage
- Welfare and structural pricing implications

Open questions for PhD research

How does memory aggregate? Do institutions remember differently? Can memory-based measures *predict* returns out of sample? These are first-order questions — and largely open.

Takeaways

1. **Memory is a new primitive.** Beliefs are built from *recalled*, not actual, experiences — and recall is cue-dependent, associative, and biased
2. **The mechanism is two-step.** Cued retrieval + simulation. Extrapolation and recency *emerge* rather than being assumed
3. **It scales.** From the lab (ESZ) to 17K investors (JLPY) to the cross-section over 55 years (Charles & Sui) to prices (Charles)
4. **Markets reflect memory.** Price overreaction in the lab is as large as belief overreaction; in the field, memory cues move prices and *reverse*
5. **Memory is upstream.** It shapes the belief input that extrapolation, sentiment, and attention then act on
6. **Rich research agenda.** Aggregation, institutions, long horizons, trauma, out-of-sample prediction — all open

Bottom line

Memory gives behavioral finance a *microfoundation* for beliefs — and a new lever for understanding prices.

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